WEEK 1 DAY 2:

Question 1: From the Groq and OpenAI router

https://console.groq.com/home

https://developers.openai.com/api/docs/models/all

--List down the Multimodels

Answer:

Multimodels in Groq

- **LLaMA 3.2 Vision** - Image + text input, Image Understanding and Visual Q&A

Multimodal Models in OpenAI

- **GPT-5** – Text + Image + Audio (+ Video in some cases), Advanced reasoning & agents
- **GPT-4o** – Text + Image + Audio, Real-time interaction & voice
- **GPT-4.1** – Text + Image, General-purpose multimodal tasks
- **GPT-4o mini** – Text + Image + Audio, Lightweight & fast applications
- **GPT Image (gpt-image-1)** – Text/Image → Image, Image generation & editing
- **GPT Realtime** – Audio + Text, Live voice conversations
- **GPT Audio** – Audio ↔ Text, Speech-to-text & text-to-speech
- **Sora** – Text → Video, Video generation

Question 2:

2) Take 3 models from each service provider

https://openrouter.ai/models

Context window, Input output token cost

Answer:

S No	Provider	Model	Context Window	Input Cost ($/M tokens)	Output Cost ($/M tokens)	Notes
1	**OpenAI**	GPT-4o	128K	5	15	Multimodal, real-time
		GPT-4.1	1M	5	15	Long context
		GPT-OSS-120B	131K	0.15	0.6	Low-cost
2	**Anthropic**	Claude Sonnet 4.6	1M	3	15	Balanced
		Claude Opus 4.6	1M	5	25	Premium
		Claude Haiku 4.5	200K	1	5	Fast
3	**Groq**	GPT-OSS-20B	131K	0.075	0.3	Cheapest
		LLaMA 4 Scout	128K	0.11	0.34	Efficient
		Kimi K2	256K	1	3	Agentic
4	**Qwen (Alibaba)**	Qwen-Turbo	1M	0.0325	0.13	Very cheap
		Qwen-Plus	131K	0.4	1.2	Balanced
		Qwen3 VL 30B	131K	~0	~0.001	Multimodal (vision)
5	**Meta (LLaMA)**	LLaMA 4 Maverick	~1M	0.15	0.6	Multimodal (image+text) (OpenRouter)
		LLaMA 3.3 70B	128K–131K	~0.10–0.12	~0.30–0.32	Strong open model (Price Per Token)
		LLaMA 3.1 8B	— (smaller)	0.02	0.05	Ultra cheap lightweight (OpenRouter)

Question 3: use the Tokenizer with different prompt and see how the tokens are generated

https://platform.openai.com/tokenizer

Answer:

S No	***Prompt***	***No of Tokens***	***No of Characters***
1	Many words map to one token, but some don't: indivisible. Unicode characters like emojis may be split into many tokens containing the underlying bytes: 🤚 Sequences of characters commonly found next to each other may be grouped together: 1234567890	53	252
2	I want to prepare a regular document explaining all the bank rules	12	66
3	Generate a image	3	16
4	Hi	1	2
5	riverbank	2	9